

Transfer Learning versus Random Initialization

Isolating feature quality from trainable capacity in scene classification

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Computer Vision · Prof. Bruce Maxwell · July 2026

Transfer learning reliably beats training from scratch, but the standard comparison moves two variables at once: the quality of the pretrained features and the number of parameters left trainable. This study adds a capacity-matched control, a randomly initialized network frozen at the same depth, so the two can be varied independently. On six-class scene classification with ResNet-18, pretrained features are worth 26.7 accuracy points once capacity is held fixed, while capacity on its own accounts for 18.

CONTENTS AND CONTEXT

Agenda and Introduction

Pretrained backbones are the default for small vision datasets. This study asks what that default is actually buying.

Agenda

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The practice

Load ImageNet weights into a standard backbone, freeze the early layers and fine-tune the tail. Near-universal for datasets below a few thousand images.

The task

Six-class natural scene classification: buildings, forest, glacier, mountain, sea and street. 14,034 training and 3,000 held-out test images on ResNet-18.

The question

The practice demonstrably works. What makes it work? Genuinely better features, or simply a smaller optimization problem than training everything?

Both explanations predict exactly the same headline number, so the comparison everyone runs cannot separate them. Doing so requires a condition that moves one while holding the other fixed.



THE CONFOUND AND THE AIMS

Problem Statement and Research Objectives

Freezing the early layers changes two things at once, so the standard comparison cannot attribute the win to either.

OBSERVED

A pretrained, partly frozen network beats one trained from scratch.



FACTOR A

Feature quality

The frozen weights already encode edges, textures and shapes learned from a million labeled images, before your data arrives.



FACTOR B

Trainable capacity

Freezing also shrinks how many parameters the optimizer can move, which helps on 14k images whatever those weights mean.

Goal. *Vary one at a time.*

Research objectives

- 01** Quantify feature quality, capacity fixed
Pretrained-Frozen against the capacity-matched control.
- 02** Quantify the contribution of capacity
The frozen random control against a fully trainable network.
- 03** Locate where fine-tuning stops paying
A freeze-depth sweep, linear probe to full fine-tuning.
- 04** Measure the first layer's reach
Replace conv1 with a fixed Gabor bank, then unfreeze upward.



PRIOR WORK

Literature Review

Four established findings frame the study; one gap motivates it.

Pretrained features transfer off the shelf

Razavian et al., CVPR-W 2014 · Kornblith et al., CVPR 2019

A frozen backbone plus a linear classifier is a strong baseline across tasks, and ImageNet accuracy tracks transfer accuracy closely.

Transferability is layer-dependent

Yosinski et al., NeurIPS 2014

Early layers learn generic edge and texture filters that transfer widely; later layers specialize to the source task.

Random weights are a legitimate baseline

Jarrett et al., ICCV 2009 · Saxe et al., ICML 2011

Untrained convolutional filters still yield usable representations, because some architectures are frequency selective by construction.

Transfer gains can be over-parameterization

Raghu et al., NeurIPS 2019

In medical imaging much of the apparent benefit is attributable to model size rather than to genuinely reused features.

Gap. *No prior study holds trainable capacity fixed while varying only the quality of the frozen features on a single architecture.*



Methodology

Three conditions on one architecture. Only the treatment of the early block θ_f changes; everything else is held constant.

CONDITION	EARLY BLOCK θ_f	LATE BLOCK θ_t	TRAINABLE SHARE
Pretrained-Frozen	ImageNet weights, frozen	trained · 8,396,806	75%
Random-Frozen <i>control</i>	random weights, frozen	trained · 8,396,806	75%
Random-Full	random weights, trained	trained · 11,179,590	100%

PT-F vs R-F isolates feature quality, capacity matched R-F vs R-FL isolates trainable capacity, both random

conv1 + bn1	layer1	layer2	layer3	layer4	fc (6-class)
θ_f frozen · 2,782,784 · BN in eval()				θ_t trainable · 8,396,806	

requires_grad = False does not freeze BatchNorm: running statistics keep updating on every forward pass, so a block believed to be frozen quietly drifts. Both frozen conditions hold BatchNorm in eval mode; Random-Full updates normally, by design.

Protocol and setup

- Stratified 90/10 split · 12,631 / 1,403
- Train 3 conditions × 3 seeds (42, 100, 2026)
- SGD 1e-3, mom 0.9, wd 1e-4, batch 32
- Early stop on val loss, patience 5
- Evaluate once on 3,000 test images
- Report mean ± std across seeds



RESULTS

Results I: Accuracy, Cost and Convergence

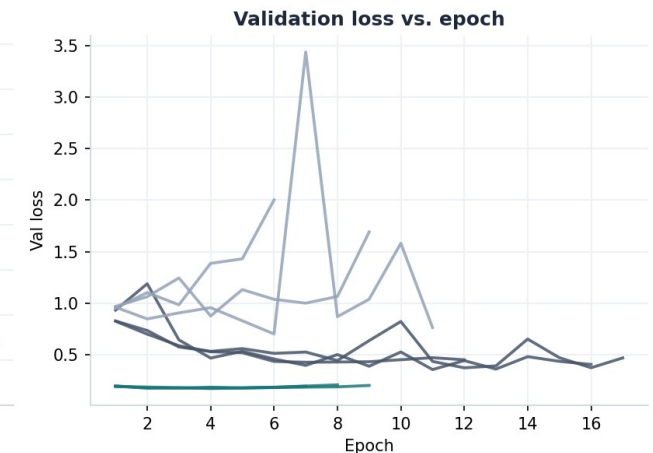
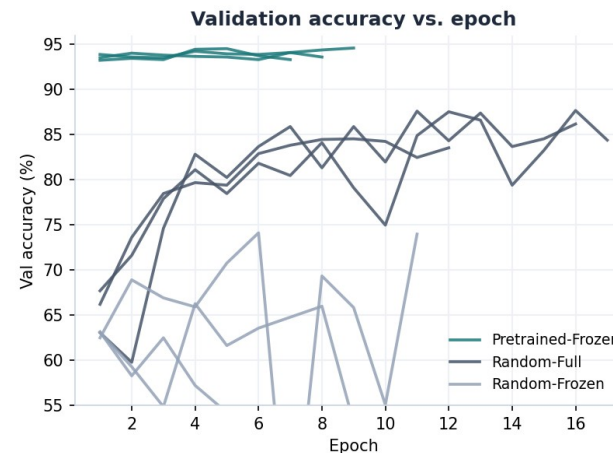
Capacity-matched, pretraining is worth 26.7 points, at a quarter of the wall clock. Mean of three seeds.

TEST ACCURACY



CONVERGENCE AND COST

Pretrained-Frozen	8.0 epochs	103 s
Random-Frozen	8.7 epochs	113 s
Random-Full	15.0 epochs	426 s



Pretrained-Frozen is above 93% at epoch one and flat by epoch four. Random-Full climbs for 12 to 17 epochs to a lower plateau, and Random-Frozen never settles: the optimizer has capacity but nothing useful to work with.

PT-F vs R-F: +26.7 points. *Identical trainable parameters, data and recipe; the only difference is whether the frozen weights came from ImageNet. R-F to R-FL recovers 18 points through capacity alone, and still finishes nine points short.*

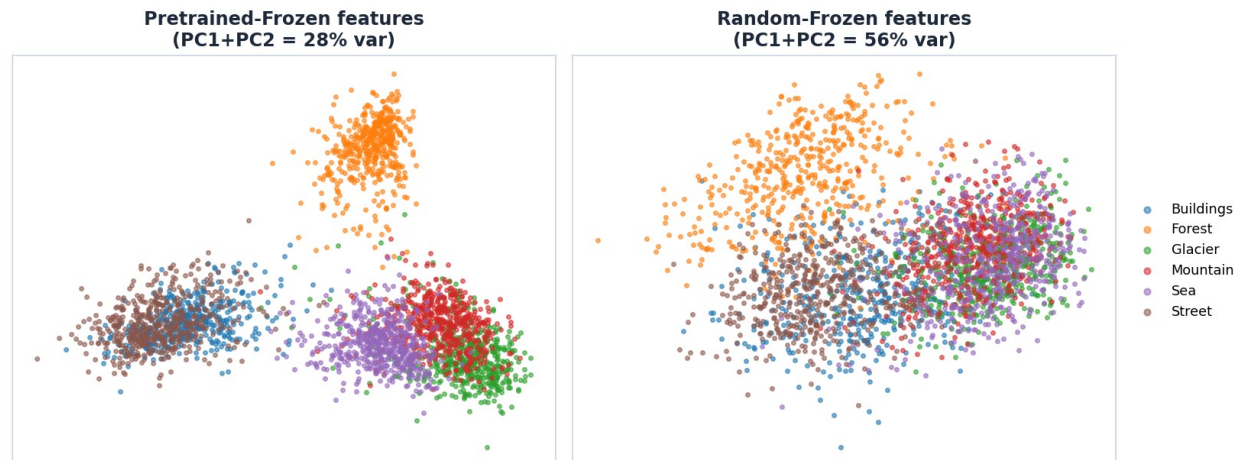


RESULTS

Results II: Feature Space and Error Analysis

The class overlaps in the pretrained feature space are exactly the confusions the classifier makes.

PCA of penultimate (512-d) features on the test split · seed 42



PCA of the 512-d penultimate activations, test split, seed 42. Pretrained features fall into clean clusters before the classifier acts on them; the random frozen features stay entangled apart from forest.

PRETRAINED-FROZEN CONFUSION · SEED 100

	B	F	G	M	Se	St
B	408	0	0	0	1	28
F	0	471	0	1	0	2
G	1	4	458	71	16	3
M	1	2	28	484	9	1
Se	2	0	2	3	503	0
St	26	0	0	2	0	473

rows true, columns predicted

Glacier ↔ Mountain · 71 and 28

Shared snow, ridges and rock texture.

Buildings ↔ Street · about 27 each way

Streets contain buildings, and the reverse.

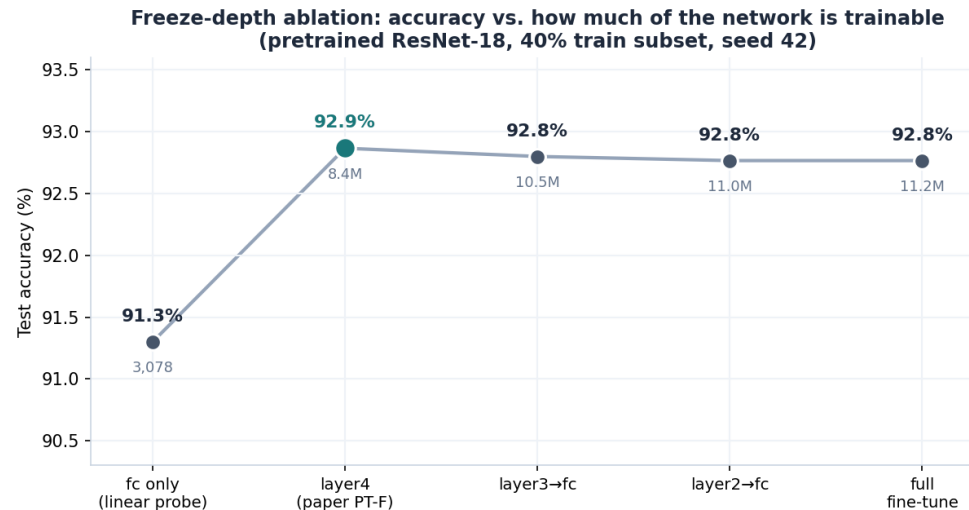
Same pairs in every condition

Larger under random init, structurally identical.

RESULTS

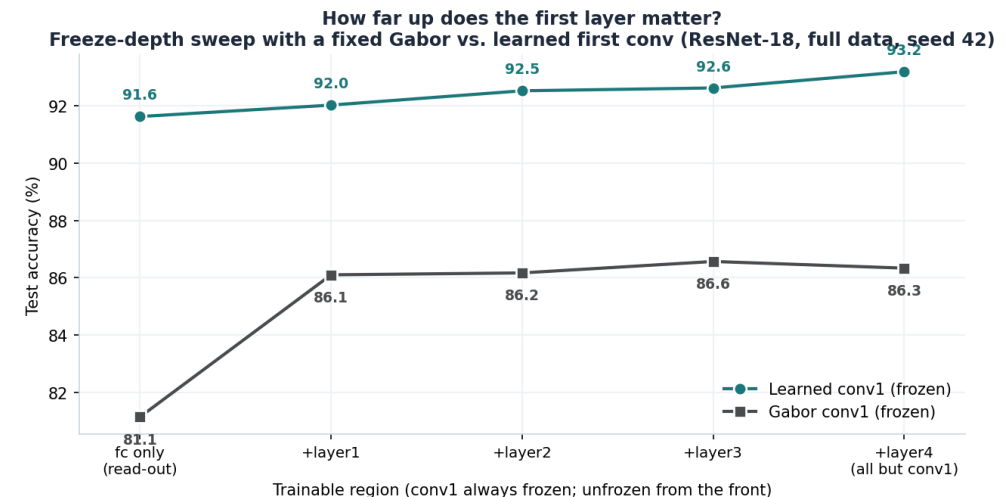
Results III: Ablation and Extension

Most of pretraining's value needs almost no fine-tuning, and a fixed filter bank cannot reconstruct the learned first layer.



Freeze depth · 40% training subset, seed 42

A bare linear probe on frozen features already reaches 91.3% from 3,078 trainable parameters. Unfreezing layer4 adds 1.6 points. layer3, layer2 and full fine-tuning are all flat at 92.8% with up to 3.6× the parameters, so the elbow is exactly at layer4.



Fixed Gabor conv1 · full data, seed 42

conv1 swapped for a fixed bank of 8 orientations × 4 frequencies × 2 phases. The read-out sits 10 points low; unfreezing layer1 alone returns 5 of them and layer2 to layer4 add nothing. Seven points never return.

INTERPRETATION AND SCOPE

Discussion and Limitations

What the two clean comparisons establish, and what this design does not.

Discussion

Feature quality outweighs capacity

26.7 points from pretrained features against 18 from full capacity. Both matter; the representation matters considerably more.

The value sits in the generic early layers

A linear probe already reaches 91.3% and unfreezing past layer4 adds nothing, matching the layer-dependent transferability of Yosinski et al.

Learned first-layer filters are not reconstructible

A 7-point Gabor gap that full downstream retraining cannot close qualifies the random-filter results of Jarrett et al. and Saxe et al. at depth.

Practical consequence

On a small dataset with a tight compute budget, freeze deep and fine-tune little.

Limitations

One dataset, one architecture

The 26.7-point figure is not claimed to generalize across domains.

Three seeds, no significance test

Gaps far exceed the within-condition spread, but no test is run.

No data augmentation

Omitted deliberately; absolute accuracies sit below tuned baselines.

Ablation on a 40% subset

Run on reduced data for tractability. Read the shape, not the values.

Gabor bank untuned

The 7-point residual is therefore an upper bound on the gap.

The control is not a method

Random-Frozen holds capacity fixed. It is not a way to train anything.



FINDINGS AND NEXT STEPS

Conclusion and Future Work

What the capacity-matched design establishes, and the four extensions it makes straightforward.

Conclusion

- i.* **Pretraining is feature quality, not parameter count**
Capacity-matched, pretrained features are worth +26.7 points over random ones.
- ii.* **Capacity helps, and cannot close the gap**
Full capacity from scratch reaches 84.7%: nine points short, at four times the compute.
- iii.* **For small data, transfer learning wins outright**
Best accuracy and fastest convergence, and the ablation says minimal fine-tuning.

The random-frozen control is what makes the first point a measurement rather than an impression.

Future work

Test the ratio off-distribution

Repeat the comparison on a domain far from ImageNet, where Raghu et al. found size dominating.

Separate depth from capacity

Replace the random control with a parameter-matched shallow network.

Scale the pretraining set

Sweep pretraining corpus size: does the 26.7-point gap grow or saturate?

Push the reconstruction limit

Tune the bank and try other fixed bases. Is the gap specific to Gabor?



CREDITS AND BIBLIOGRAPHY

Acknowledgments and References

Peer-reviewed venues only.

Acknowledgments

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Intel Image Classification dataset	Natural scene imagery across six categories, distributed publicly via Kaggle.
PyTorch and torchvision	ResNet-18 implementation and the ImageNet-pretrained weights used throughout.

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QUESTIONS

Thank you

Capacity-matched, pretrained features are worth 26.7 accuracy points. The control that establishes it is a model nobody would ever deploy.

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